

Explain It Like You Built It

What is Astro and how does it work? Why do all files have .astro and not the familiar .html.

We can think of Astro as a factory which means before we get a product it is made somewhere with a pipeline that is invisible to the end user. Now what does this factory look like for our static website. Our factory is Astro and the main reason we used it is because it allows us to compartmentalize our website into pieces that get assembled into the HTML that get's served to the end user once deployed. How does this work?

Let's look into the pipeline for specifically the projects part of the website to understand how it works. We have our data in typescript (.ts file) and this contains the actual projects data for every project we are showcase it has information like summary, image and title.

This piece is fed to the next piece of the line which is the pages part which allows us to have one dynamic page [slug].astro that will create 5 different pages for each of the projects in the data. It uses other pieces like the BaseLayout which all pages build upon and CaseStudyLayout which all project pages build upon and is able to create a page for each project.

The page passes its project data down to the components as props (Astro's term for the data a component receives), and the components render inside the page.

More specifically, the page already holds the data it imported. It passes that data to the components as props, the components render themselves in place using it, and Astro assembles the whole thing into the final HTML.

Now to answer the question of the file extension .astro. This is what identifies a file as having the unique Astro blend of typescript at the top and html all in one file (the typescript is read first) and it allows the combining of pieces in one file instead of a html file with a separate .js for example.

One important note about this factory is that all of this assembly happens at build time not during live visits when a visitor opens the website. All pages for the static website are done and completed before it is deployed and the full HTML is what gets rendered.